Weekly Review
R213 - Functional Programming

Complete the weekly review, you'll have 15 minutes.

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R213 - Functional Programming

Mark that review!

Once you've done it:

1. Submit the score into the Google Form.
2. Update your weekly review tracker with **targets**.

Starter

In **2024**, how many **every minute**?

1. TikTok videos uploaded
2. Snaps sent
3. Microsoft Teams **minutes** recorded
4. Emails sent

Starter

In **2024**, how many **every minute**?

1. TikTok videos uploaded
2. Snaps sent
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Starter

In **2024**, how many **ever**

1. TikTok videos upload
2. Snaps sent
3. Microsoft Teams **minutes** recorded
4. Emails sent

How much do you think was spent on AI **every minute** in 2024?

What about 2025?

Researching the numbers

Go to `statista.com` and search ‘Big Data’.

Find the topic `Big data - statistics & facts`

Have a look for:

1. How has the amount of data changed over the years?
2. How has the revenue for big data change?
3. How has AI uptake changed?

Why?

Employability Skills

Data analysis is a very important ‘soft skill’ which is vital for many jobs, even if it isn’t in the name!

- Mangagement Consultant
- Actuary / Insurance Broker
- Quantative Analyst

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Data is often used to make decisions, which leads to problem solving, debate and discussion.

Big Data, on Spec

Content	Additional Information
Know that ‘Big Data’ is a catch-all term for data that won’t fit the usual containers. Big Data can be described in terms of: • volume - too big to fit into a single server • velocity - streaming data, milliseconds to seconds to respond • variety - data in many forms such as structured, unstructured, text, multimedia.	Whilst its size receives all the attention, the most difficult aspect of Big Data really involves its lack of structure. This lack of structure poses challenges because: • analysing the data is made significantly more difficult • relational databases are not appropriate because they require the data to fit into a row-and-column format. Machine learning techniques are needed to discern patterns in the data and to extract useful information.

	‘Big’ is a relative term, but size impacts when the data doesn’t fit onto a single server because relational databases don’t scale well across multiple machines. Data from networked sensors, smartphones, video surveillance, mouse clicks etc are continuously streamed.
Know that when data sizes are so big as not to fit on to a single server: • the processing must be distributed across more than one machine • functional programming is a solution, because it makes it easier to write correct and efficient distributed code. Know what features of functional programming make it easier to write: • correct code	Functional programming languages support: • immutable data structures • statelessness • higher-order functions.

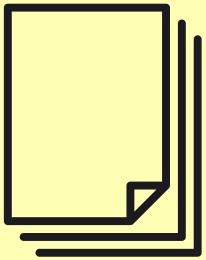
• code that can be distributed to run across more than one server.	
Be familiar with the: • fact-based model for representing data • graph schema for capturing the structure of the dataset • nodes, edges and properties in graph schema.	Each fact within a fact-based model captures a single piece of information.

The 3 Vs of Big Data

On your whiteboard, write down the 3 Vs of Big Data.

For each, write a definition of what this term means for Big Data.

The 3 Vs of Big Data



Volume

The size of the data to be processed.

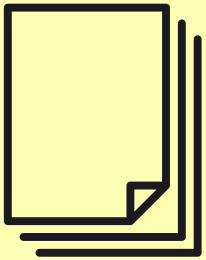
If it is too large to be processed as a **single dataset** then it is considered ‘big’.

The 3 Vs of Big Data

Velocity

The speed at which data is generated and to be processed.

The 3 Vs of Big Data

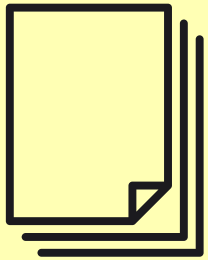


Variety

Refers to all the structured and unstructured data.

Variety is all about the ability to classify the incoming data into various categories.

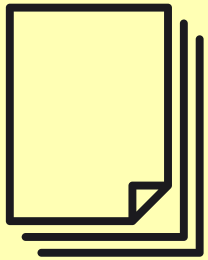
Variety



Data can appear in many forms:

- Structured Data
  - Represented in tabular form so can be easily modelled in a relational database.

Variety



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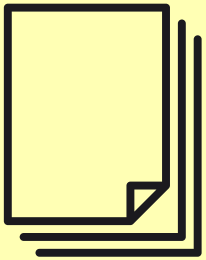
- Semi-structured data
  - data with variable structure. Eg XML or JSON formatted files. Although a structure can be applied, it is not always consistent.

Variety



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  - Represented in tabular form so can be easily modelled in a relational database.
- Semi-structured data
  - data with variable structure. Eg XML or JSON formatted files. Although a structure can be applied, it is not always consistent.
- Unstructured Data
  - Context variable so can not be modelled in advance and can not be fitted in a relational database or tags. Eg video, images, email messages etc.

Recall

What features of functional programming make it ideal for Big Data?

Recall

What features of functional programming make it ideal for Big Data?

Easier to write correct code

- Because it is very strictly defined programming paradigm

Immutable data structures

- A data structure whose state can not be changed after it is created. There are no side-effects. These are inherently ‘thread-safe’.

Recall

What features of functional programming make it ideal for Big Data?

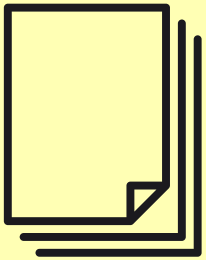
Statelessness

- Functional programs are entirely predictable. Same inputs always result in the same output, no matter how often they are run, or in which order they are run.

High-order functions

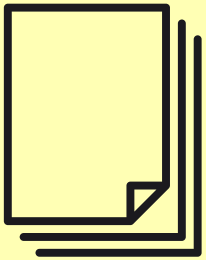
- A function can be passed as an argument to another function.
  Eg reduce (map(listofthings)). These can be easily distributed across multiple data servers to process large quantities of data.

Fact Based Model



- A conceptual model for modelling data.
- Each fact in a fact based model captures a single piece of information.
- Data is immutable – it cannot be altered except to delete the data – completely different to relational model.
- Any change is recorded as a new fact NOT an update. The dataset continually grows with the addition of immutable timestamped data.
- This minimises the impact of human error. If bad data is written, earlier good data still exists.

Fact Based Model



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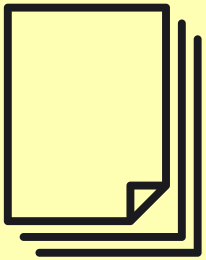
An example of a Graph Schema for a Fact Based Model

Graph Schema



This captures the structure of a dataset stored using a fact based model.

Nodes

- Entities of the system. EG. student or webpage

Edges

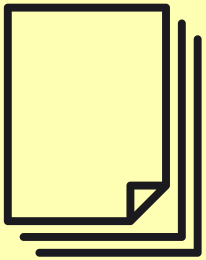
- Relationships between nodes. EG. a student has visited that page.

Properties

- Information about entities. EG. surname or total views.

Solid lines link nodes, **dashed lines** link properties to nodes.

The big Big Data Summary



What is Big Data?	3V's – Volume, velocity, variety.
What does it look like?	Its made up of 'facts'. It isn't always a nice table like a database (variety – structured, semi-structured, unstructured).
How can we lay it out?	Diagram with nodes for entities and lines that link entities and properties
How do we deal with it?	Use Functional Programming!
What makes FP so good?	Immutable Data Structures, Statelessness, Higher-Order Functions.

Big Data Questions based on pre-work

On Google Classroom:

L228 - Big Data Questions